

SAFE WORK METHOD STATEMENT

Persons responsible for supervising, inspecting work, work methods, plant & equipment and tools	Team Leader (Supervisor)	Date:	
High Risk Construction Work:	Vacuum Suction Excavator (RSP) – Safe Operation	Project:	
SWMS prepared Patrick Monaghan - Paneltec Group - Operations Manager and in consultation with management and workers – 2025. SWMS Review Date: August 2026 unless required earlier		Address:	
Signature: _____			
Approval to use this SWMS: “I certify that the attached SWMS has been reviewed by me, is endorsed for compliance with WorkSafe guidelines and is approved for use”			
<u>John Guy</u> Name	<u>Director - Paneltec Pty Ltd</u> Position	<u>0408 449 023</u> Phone	<u>31/08/2025 – V12.0</u> Date
		<u>[Signature]</u> Signature	

RISK MATRIX

Level	Description of Consequence or Impact	Consequence	Likelihood / Probability		
			L Likely	M Moderate	U Unlikely
H (1) (High level of harm)	Potential death, permanent disability or major structural failure/damage. Off-site environmental discharge/release not contained and significant long-term environmental harm.	H (1) (High)	1	1	2
M (2) (Medium level of harm)	Potential temporary disability or minor structural failure/damage. On-site environmental discharge/release contained, minor remediation required, short-term environmental harm.	M (2) (Medium)	1	2	3
L (3) (Low level of harm)	Incident that has the potential to cause persons to require first aid. On-site environmental discharge/release immediately contained, minor level clean up with no short-term environmental harm.	L (3) (Low)	2	3	3
Level	Likelihood / Probability				
Likely	Could happen frequently				
Moderate	Could happen occasionally				
Unlikely	May occur only in exceptional circumstances				

HIERARCHY OF RISK CONTROL

The hierarchy of controls must be applied in the order below for all identified hazards.

Level 1 Eliminate the hazard
Level 2 Substitute the hazard with something safer.
Level 3 Isolate the hazard from people. Reduce the risks through engineering controls.
Level 4 Reduce exposure to the hazard using administrative actions.
Level 5 Use personal protective equipment
The control measures that you put in place should be reviewed regularly to make sure they work as planned

ACTIVITY ANALYSIS

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Residual Risk Class
1.0 General Safety Requirements  	Personal injury to operator and other workers.	1	1. All workers are to wear required PPE in accordance with Paneltec's Pipe Policy: • Highly Visible Workwear – Long Pants & Shirt (Sleeves Down) • Safety Boots • Hard Hat – No caps underneath • Safety Glasses - AS/NZS1337 • Prescription Safety Glasses (Where Applicable) • Hearing Protection (Where Required) • Face Shield in addition to Safety Glasses when cutting, grinding, using compressed air or <u>high-pressure water jetting</u> • Construction Gloves Cut Factor >4 & Glove Clip	All	3
	Risk of Striking underground services – resulting in Electrocutation, asphyxiation or engulfment	1	2. Review BYDA Pack - Conduct tailgate meeting with all persons on-site, review DBYD location information, ensure all plans are available that are listed as sequence numbers on the BYDA cover page. Ensure each have been reviewed.	Operators	2
		1	3. Underground Asset Locations - Ensure all underground assets to be potholed have been located by a Certified BYDA Locator and the depth is available for reference.	Operators	3
		1	4. Site Specific Risk Assessment/Job Safety Environmental Analysis SSRA-JSEA - Conduct Paneltec SSRA-JSEA on electronic smart device, use a hardcopy if mobile range is not available.	Operators	3
		1	5. Complete Paneltec Excavation Permit & Checklist (All Sections).	Operators	2

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Residual Risk Class
2.0 Set out work area.  	Struck by Vehicle or construction plant.	1	1. Barricade the work area, to prevent inadvertent access by public. 2. Implement relevant Traffic Guidance Scheme (TGS) and ensure pedestrians have a designated, safe passage around the worksite. 3. Ensure all operational flashing lights and reversing beepers on all plant are functional. 4. Implement Traffic Guidance Scheme (TGS) and Traffic Management Plan (TMP) NOTE: When working in higher risk areas additional traffic controllers may be required to implement more complex TGS's – Consult Management immediately if the need is identified 5. Implement <i>DANGER- Compressed Air, High Velocity Vacuum & Water Jetting Signs</i> at either end of the cordoned off work area 6. Position spray shields adjacent to the hole to be excavated	Operator Operator/Traffic Controller Supervisor/Operator /Traffic Controller Operator	2
3.0 Pre-start checks - Vacuum Suction Excavator 	Unsafe machine potential for injury to operator or bystanders	1	1. Pre-start checks are to be completed prior to operation and are to be submitted on our WHS monitor platform via scanning the QR Code. 2. All safety devices: emergency stops, reversing alarms, rotating flashing lights, communication device (radio), seat belts and fire extinguisher are effective and operational. 3. Mobile plant operators hold the appropriate licenses and competencies for the plant they are required to drive and or operate. 4. Mobile plant and vehicle operators always find a stationary position in a safe place prior to handling a mobile phone. 5. Safety related deficiencies are resolved before equipment is put into operation. 6. Mobile plant is always switched off and braking mechanisms are applied before being left unoccupied. Chocks used when parked on an incline. 7. The remote-controlled operation of the truck shall only be utilised where necessary and be in accordance with the specified manufacturer's instructions.	Operator	2

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Residual Risk Class
4.0 Electrical/Gas Hazards 	Electrocution Explosion Asphyxiation	1	1. Conduct visual inspection for signs of underground gas or electrical infrastructure in the project area. Visual Indicators include, but may not be limited to: NATURAL GAS: Gas markers on kerb, gas meters on houses, asphalt scars, gas footpath isolation valves. ELECTRICAL: Green electrical turrets, armored black wire (XLPE) heading underground from power poles, buried cable signs), orange conduit coming up from the ground	Supervisor/Operator	2
		1	2. Conduct a visual inspection for overhead electrical infrastructure: If personnel are authorised persons (Having completed <i>UETTDREL14 Working safely Near Live Electrical Apparatus as a non-electrical worker</i>) within the last 5 years, the following or the workers working in the vicinity of overhead electric lines are supervised by a person meeting the above training requirements can work within the following limits of approach: 1 m of positively identified low voltage conductors 240 V and telecommunications cables. 3m from High Voltage power lines 11KV to 33 KV 8 m from Extra High Voltage 110 KV to 220KV NOTE: Unauthorised persons are to maintain approach distances of 6.4m from LV & HV. From EHV the clearance is 10M If the distance between the vehicle/boom or operating implement encroach on the clearances above, control measures must be implemented to reinstate an adequate clearance. Eg Extension hose ect	Supervisor/Operator	2
		1	3. Ensure a spotter is always in place when maneuvering vehicle/boom/cassette or operating implement around the overhead electrical safety envelope. Spotter is to be authorised/trained.	Supervisor/Operator	2

			- In instances where the boom needs to be locked in place to avoid operating at a certain height, this can be facilitated by locking the boom in place at its maximum required operating level by setting the adjustment on the boom height and radial reach locking slide. - If gas or electrical indicators are known, refer to/conduct DBYD enquiry and review TasGas Networks & TasNetworks Dial Before You Digs - Ensure Paneltec's Emergency Procedures are available for reference in the event of an emergency.	Operator Supervisor Operator Operator	
8.0 Risk of Fall – Working at Heights 	Fall from height – Death, paralyzation or injury to operator	1	- In all instances where access to the top of the truck is required on the project site, the truck must be isolated, in gear, breaking mechanisms applied, and the wheels chocked. A safety harness must be worn and a self-retracting lanyard must be affixed to the restraint points on top of the truck. - If the work is on a TasWater site, a permit must be requested and authorised. Once the permit is obtained work can proceed complying with the requirements of the Working at Heights Permit & Rescue Plan.	Operator Supervisor	3
8.0 Operating the Vacuum Suction Excavator – Jetting & Lancing 	Injury to operator or bystanders – high pressure- dirt and stones can be flicked around.	1 1	- All workers are to wear required PPE in accordance with Paneltec Pipe Policy: - Highly Visible Workwear – Long Pants & Shirt (Sleeves Down) - Safety Boots - Hard Hat – No caps underneath - Safety Glasses - AS/NZS1337 - Hearing Protection (Where Required) - Face Shield in addition to Safety Glasses when cutting, grinding, using compressed air or <u>high-pressure water jetting</u> - Construction Gloves Cut Factor >4 & Glove Clip - Keep operating area clear of public access Safety area: Area around the suction excavator that can only be entered by operating personnel.	Supervisor Operator All Operator	3 2

		1 2	4. Emptying area: Tipping area of the material container. Nobody can be in this area when the material container is tipped – Spotter to Police Area 5. Around the suction excavator, there must be a space of 1 m when it is operating (safety margin). The turning radius for maneuvering must also be considered for this space 6. Stay alert and beware of the force of the water or air pressure coming out the end of the lance and when using the vacuum hose be aware of the suction pressure. 7. Do not overreach or stand on unstable supports. 8. Never aim the water jet or air lance in the direction of persons or animals as the water jet and compressed air comes out of the nozzle at high speed and with high pressure. 9. Do not leave the operating vacuum unattended. If the vacuum is not required switch off. 10. Follow directions as per the operations manual. 11. There are at least two emergency stop buttons on the vehicle: • On the remote controllers • On the control panel of the working headlight • And on the manual emergency radio transmitter • Ensure all workers are familiar with their whereabouts	Operator Operator Operator	3 3
9.0 RSP remote steering system OPERATION OF MOBILE PLANT 	Truck run/roll away, rollover, damage to property – Crush injuries to workers and or public, damage to cars and or infrastructure.	1	The RSP remote steering system is for automatic steering of vehicles in construction site areas only. Sensors fitted to either end of the truck ensure no persons or objects are hit by the truck whilst in remote steering mode. If the sensors detect an object, the truck will divert back to park with the handbrake applied. The handbrake will be in the on position on activation and is overridden by the trucks computer and hydraulic system whilst in communication with the remote. The RSP steering system is conceived as replaceable equipment for the final positioning of the suction excavator in the construction site area (steering) in conjunction with a special control.		2

			The following must be considered if use of the RSP steering system is planned: 1. The use of the RSP steering system in areas at risk of explosion <u>is not permitted.</u> 2. The mechanical interfaces are formed by two catches that go into the steering wheel and a bar with a locking system as a torque converter bar; 3. The RSP steering system must only be operated in closed cabs and must be protected from weather influences such as rain and snow. 4. Before putting the RSP steering system into operation, it must be ensured that there are no dangers that could be triggered because of the automatically steered vehicle. 5. The operator must only work with the RSP steering system if they are convinced that there are no unauthorised personnel in the danger zone 6. The operator must always ensure that there is sufficient visibility in the danger zone and working area. The visibility in this area must not be impaired at any time. 7. Therefore, in the event of bad lighting conditions, the operator of the RSP steering system must always ensure sufficient lighting or stop operation. <u>Forbidden</u> applications of the RSP steering system: • Use of the steering system outside of the construction site, • Use of the steering system in a school, playground or an area near children • Attachment of the steering system to a different type of steering wheel or vehicle • Combination of the steering system with a different controller	Operator	2
		1		Operator	
		1		Operator	2

10.0 Hose blockage	Fall from heights, Entrapped hand, leg or foot Manual handling,	1	1. Ideally all blockage removals should be undertaken at the Paneltec depot where possible or practical with use of the height gantry system	Operator	3
		1	2. If an on-site blockage occurs see 8.0 above and adorn your harness and equip your Self Retracting Lanyard (SRL)	Operator	3
			3. Ensure the vacuum fan and truck are isolated prior to unblocking any blockages.	Operator	
			4. The main point for potential blockages is the Boom Inlet Access Hatch on top of the passenger rear side of the truck.	Operator	
11.0 Operation of Air Excavation Lance	Incorrect Operation – Electrocution, Injury to operator or workers	2	5. Access to the hatch is by the vertical ladder at the rear of the truck, harness and SRL are to be used whilst climbing the ladder	Operator	3
			6. Once on top of the truck, change the SRL anchor point to one on top of the truck	Operator	
			7. Move over to the hatch, open it and check for rocks in the boom inlet to the tank	Operator	
			8. At no point shall an operator or their body parts be within 1m end of the vacuum hose whilst it is in operation due to its sheer velocity	Operator	
11.0 Operation of Air Excavation Lance	Incorrect Operation – Electrocution, Injury to operator or workers	2	9. If the boom inlet is clear and the obstruction is in the hose, the hose can be tapped with a rubber mallet whilst the unit is on	Operator	3
			10. Ensure no one is near the end of the hose whilst conducting vacuum hose mallet tapping	Operator	
			1. Operation of the air excavation lance shall only be by trained and competent operators wearing all PPE identified in 1.1 Above	Operator	
			2. DO NOT use compressed air from the lance to:	Operator	
11.0 Operation of Air Excavation Lance	Incorrect Operation – Electrocution, Injury to operator or workers	2	• Transfer flammable liquids. Static electricity build-up can discharge and ignite the liquid.	Operator	3
			• Empty containers. The container could rupture due to excessive internal pressure.	Operator	
			• Clean clothes, hair, or skin or point the lance at anyone.	Operator	
			3. The Air lance is constructed of electrical non-conductible fibers, however contact with any underground infrastructure should be avoided	Operator	
11.0 Operation of Air Excavation Lance	Incorrect Operation – Electrocution, Injury to operator or workers	2	4. The lance can be used to give pulse type air burst to the excavation area with intermittent blasts – useful in built up or crowded areas	Operator	3
			5. The lance can also expand the surrounding soil with an extended pulse charge when driven into the sub soil of the excavation area, which helps loosen soil, clay and rock lodgments. This shall only be used with a 4m radial (8m Diameter) safety zone surrounding the imbedded air lance.	Operator	
				Operator	
				Operator	

12.0 Operation of Air Powered Utility Saw	Contact of the saw bar with operator – Injuries to operator	1	1. Prior to using the air powered utility saw, please ensure the Trained and experienced operator adorns all PPE outlined in 1.0 Above. 2. The decision to use the utility saw is the responsibility of the RSP operator 3. The water fitted saw can be used to cut Ductile/Iron/Galv FE or PE pipes Please inspect the utility saw prior to its use: • All air supply hoses are in good condition • Chain in good condition • Saw Chain - rotation good, sharp, no broken chain links/teeth? • Stop/Safety Switch - in working order? • Guards – In place not split or damaged?		2
	Incorrect use of Utility Saw – Injury to operator				
	Pre-Use Inspection – Faulty Equipment – Injury to Operator				
	Cutting – Injury to workers or persons in area-	1	Preparing to Cut 1. To reduce the risk of kickbacks, keep a firm grip on the saw with your thumb underneath the handhold and avoid touching objects with the tip of the bar. 2. Obtain stable footing before cutting 3. Cut at shoulder height or below, whenever possible 4. Avoid contacting other objects with the backside or The nose of the chain (excavation, soil ect)		2
	Cuts Manual handling sprains and strains Noise (hearing loss) Vibration (circulatory damage) Electricity (electric shock) Surfaces slippery from cutting residue (falls, sprains) Dust	1	To avoid Bar Kickback when using the chainsaw, the following rules should be observed: • The chain should always be sharp • Make sure the chain oil pump is working properly • Ensure the bar tip is well oiled • Ensure Proper Chain Brake & Rakers • Avoid contact with objects		2
	Working Alone	1	Do not work alone with the Utility Saw as this can be hazardous because of the potential need to provide assistance to the operator in the event of an unsafe incident or emergency situation		

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Residual Risk Class
13.0 Operation of Air Powered Quick Cut Saw	Untrained Operator – Injury to their self or others.	1	- Prior to using the air powered Quick Cut Saw, please ensure the Trained and experienced operator adorns the following PPE: - Construction Gloves - Highly Visible long sleeved pants and work shirt - Steel toed safety boots - Hard Hat with Ear Protection and Face Shield - Hearing Protection (Class 5) - Safety Glasses - Ensure the operator understands what the hazards associated with the cutting operation are, why the risks need to be controlled and how the risks are to be controlled. - Ensure DBYD have been referenced and that location have been undertaken where applicable. - Ensure the operator understands what protective equipment is needed, why it is needed, and how it is to be fitted and used.	Management Supervisor	2
The protection of other workers and the public.	Flying debris	1	Barricade the area in accordance with 2.0 above and ensure that all workers present in the construction site have their PPE adorned in accordance with 1.0	Operator	2
The type and condition of the concrete	Obstructions or resistance in the material being cut – can cause sudden kick-back, pushback or pull-in movements of the saw.	1	- Is the material you are going to cut likely to contain reinforcing steel, electrical cable conduits, gas pipes or other underground services? - Refer 1.3 above	Supervisor	2
Using the equipment Setting Up	Cuts Manual handling sprains and strains Noise (hearing loss) Vibration	1	When setting up a cutting operation, make sure that: - The exact location of the cut or penetration is clearly marked on the work area - A trolley may be used to support the cutting machine for horizontal work at low level, so that operators do not have to work on their knees. - The cutting blade is the right size and right type for the machine	Operator	2

	(circulatory damage) Electricity (electric shock) Surfaces slippery from cutting residue (falls, sprains) Dust		- The blade is in good working condition and is free from cracks and deterioration - Appropriate barricading and warning signs are erected to prevent people entering the cutting area. - Adequate lighting (natural or artificial) is provided at the work area. - A method of collecting residue from the cutting operation is put in place in order to prevent surfaces becoming slippery.		
Before cutting operation	Cuts Manual handling sprains and strains Noise (hearing loss) Vibration (circulatory damage) Electricity (electric shock) Surfaces slippery from cutting residue (falls, sprains) Dust	1	- The general condition of the air saw should be checked by the person doing the cutting before commencing each job to ensure that the cutting tool, guards, leads and air hoses are in good order. - Make sure that equipment with defects which would endanger people is not to be used. (Tag out). Before cutting operation, check that: - The cutting speed matches the drive speed, as per the manufacturer's specification. - The shaft and flanges are clean and not damaged. - The blade fits securely over the shaft. - The shaft nut is securely tightened against the outside flange. - A blade guard is fitted and is in good working order. - The drive belt is tensioned correctly. - Adequate coolant or water is readily available for wet cutting.	Operator	2
During the cutting operation	Cuts Manual handling sprains and strains Noise (hearing loss) Vibration (circulatory damage) Electricity (electric shock)	1	During cutting operation, ensure that: - The blade guard is in the lowered position. - When starting the machine the operator and other people stand outside the path of the blade. - If the machine stalls during cutting operation, the blade is raised and the outside flange and nut are checked for tightness. - Whenever blade is raised, when resuming cutting, ensure the blade is in alignment with the previous cut. - Where possible, cut from a standing posture with the feet braced and the body balanced. (In some circumstances it may be necessary to kneel	Operator	2

	Surfaces slippery from cutting residue (falls, sprains) Dust		on one knee to enable cuts to be made close to the road. In this situation, protection for the knee may be necessary). - The length of time the operator spends in fixed postures is minimised. - Plenty of water or coolant is used to suppress dust at the point of generation. (Failure to use coolant/water could create high levels of airborne dust that contains crystalline silica in concentrations to be a significant health hazard). - The throttle lock is only be used when starting the equipment. If the throttle lock used during normal operation, then the ability to cut air velocity will be reduced increasing the potential for injury. - The equipment is stopped when changing grip to move between horizontal and vertical cuts. - The handles provided are used to support the equipment. (Supporting the equipment by the belt guard is poor work practice and should be avoided). - Where electrical leads are present on-site ensure they will not be cut during cutting operations. - Ensure the air hose is not cut during cutting operations		
Maintenance	Various	1	The equipment should be inspected and maintained regularly by a competent person.	Mechanic	2
Working Alone		1	Do not work alone with the quick cut as this can be hazardous because of the potential need to provide assistance to the operator in the event of an unsafe incident or emergency situation.	Operator	2

ADDITIONAL HAZARDS IDENTIFIED	CONTROL MEASURES FOR RISK MITIGATION

FURTHER ON SITE SCENARIOS	DOCUMENTS FOR PROJECT REFERENCE	DOCUMENT REFERENCE NUMBER
Emergencies - Gas Strike , Electrical Etc.	Paneltec Emergency Response Plan	WHS - 05
Community Relations, Complaints Etc.	TasWater Community & Stakeholder Plan (if applicable) Else, direct all enquires to Paneltec Operations Manager	ISSUED ON PROJECT WHERE REQUIRED
Environmental Controls, Incidents Etc.	TasWater or Paneltec Site Environmental Plan	REFER EM-01 – EM-014 - Paneltec IMS
Pre-Start	Paneltec Tailgate Meeting & JSEA-SSRA	WHS MONITOR - AUDR-001

Personal Protective Equipment						
All site personnel must wear the required PPE specified in the controls above, such as:						
Safety Highly Visible Vests/Workwear	Safety Boots	Sunscreen	Sun ring for Hard Hat (UV)	Sunglasses (Approved Safety)	Gloves specific to the task	White reflective overalls
Hard Hat at all Times	Safety Glasses	Hearing protection where required	Long Sleeves and Pants	Dust masks	Respiratory Helmets if and when required	Face Shield when using jetting hose

Workers may require Task Specific Training depending on activity each worker will be required to fulfill.				
All workers shall hold a current Construction Induction Card				
- TasWater CDO & TasWater Induction - Activity Induction (SWMS & SWP's) - Site specific requirements (external)	Vehicle License's (Car or Truck depending on class of vehicle or mobile plant driven)	Plant Operators license's or competency assessments where required	Competencies for Work Zone Traffic Management	
First Aid certificate training	Manual Handling	Fire Extinguisher training	Fatigue Management Compliance	

Legislation & Codes of Practice (refer to company Legislation register for applicable Australian Standards applicable to works)			
Work Health & Safety Act 2012 and Work Health & Safety Regulations 2022			
Codes of Practice and Other Guides			
Confined Spaces - 2020	Hazardous Manual Tasks - 2020	Managing Noise & Preventing Hearing Loss at Work-2020	Managing Risks of Plant in the Workplace - 2020
Construction Work - 2020	How to Manage WH&S Risks - 2018	Managing Risk of Falls at Workplaces - 2020	Managing Electrical Risks in the Workplace -2019
Excavation Work - 2020	First Aid in the Workplace - 2018	Managing Work Environment & Facilities - 2020	WHS Consultation Co-operation & Co-ordination - 2018
Welding Processes - 2021	How to Safely Remove Asbestos - 2020	Managing Risk of Hazardous Chemicals - 2018	Safe Design of Structures - 2018
Guide for Managing Risks from High Pressure Water Jetting – Safe Work Aust 2020			

Plant & Equipment for this activity
- All plant is inspected prior to sending out to site - Daily pre-start checks are conducted on all plant and trucks by the operators and daily checklist filled out. - All maintenance issues addressed prior to departure. - Regular servicing as per manufactures specifications. Service History available on request to the office.

List of plant for this Activity
Scania G450 – RSP - Vacuum Suction Excavator Truck
Engineering details/certificates/regulatory approvals/permits/electrical isolation (if applicable):

Implementation, Monitoring and Review of SWMS

The Supervisor on site has the responsibility to:

- Brief each team member on this SWMS before commencing work.
- Ensure control measures have been implemented prior to the start of works.
- Observe work being carried out.
- If controls documented in SWMS are not adequate, stop the work, review the SWMS, adjust as required and re-brief the team.
- Ensure team knows that work is to immediately stop if the SWMS is not being followed.
- Retain this SWMS on site for the duration of the high-risk construction work.
- All persons involved in these particular tasks on site must sign this document as read and understood.

SWMS attendance sheet — I have been given the opportunity to provide input into the development and review of this SWMS

Name (please print clearly)	Signature	Company	Date

Daily pre-start meetings will be held to ensure all workers are informed of control measures and additional safety hazards & controls to be noted on toolbox form.